$$\frac{E_v}{N+Z} = -A + B\frac{(N-Z)^2}{(N+Z)^2}$$

where E_v is that part of the total binding energy which originates in the nuclear field, and which is proportional to the volume. A and B are constants. $E_v/(N + Z)$, the binding energy per particle, is therefore equal to a constant—A when $N = Z$, i.e. when the number of protons is equal to the number of neutrons. But if N and Z are not equal to each other, small divergences occur and in that case the simplest symmetrical function of N and Z is the expression $(N - Z)^2$. But since nuclear matter is homogeneous, the binding energy per particle can depend only on the N/Z ratio, which latter is obtained through division by $(N + Z)^2$. The most general symmetrical function would be, obviously, more complicated. But if it is developed, according to ascending powers of $(N - Z)$, in a Taylor series and broken off after the second term, we obtain exactly the equation given above. This approximation is sufficient, since in the cases of the nuclei observed, N and Z never differ much from each other.

The first term of this equation is negative, as it must be in the case of a binding energy. On the other hand, the second term, which is considerably smaller, is positive, and therefore produces a reduction of the absolute magnitude of the negative binding energy. The latter decreases when N and Z are different from each other. Therefore, from the point of view of energy economy, the most favourable situation seems to be present when a nucleus contains exactly as many protons as neutrons.

But here we must mention a correction depending on the surface tension. The particles situated on the surface are bound together less tightly than those in the interior, and this circumstance, consequently, calls for the addition of a further positive term on the right-hand side of our equation. For since the principal term, $-A$, is negative, this side of the equation is essentially negative, and something positive must be added to it in order to decrease its value. The change in the quantity of binding energy due to surface tension is proportional, in every instance, to the number of the particles situated on the surface, and hence to the surface area. But the latter is proportional to the nuclear volume (or to the total number of particles) raised to the power of 2/3his portion of the binding energy can therefore be expressed by the equation $E_0 = C(N + Z)^{2/3}$. The portion per particle is obtained by division by $N + Z$, i.e,:

$$\frac{E_0}{N+Z} = C(N+Z)^{-1/3}$$